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Why ransomware fears EDR 7.0

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Endpoint Security Overview

End Point Protection History

- **1949–1980 period (pre-antivirus days)**
  - **1949 -***Theory of self-reproducing automata*
  - **1971 – Creeper Virus** considered first known virus DEC PDP-10
- **1980s**
  - Viruses go mainstream
  - **1987 –** McAfee starts as well as few other companies in Europe
- **1990s**
  - Anti-Virus Industry is created, lots of companies start
  - All signature based approach
- **2012**
  - Cylance releases “Next-Gen AV” based on machine learning
- **2014**
  - CrowdStrike Falcon is released, just Threat Hunting, no blocking or prevention

Source: Wikipedia

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End Point Protection Methods

1) Signature based does not work any more. It's not the 90's bro.
   a) Still required in some RFPs, very limited effectiveness in modern attacks.
   b) Old habits die hard

2) Machine Learning/NGAV is a better approach
   a) Detects polymorphic viruses and other obfuscation methods
   b) Still cannot detect Zero-Days

3) Endpoint Detection and Response
   a) NGAV with Threat Hunting, ingest more data, catch more stuff right?
   b) Catching Zero Days and Ransomware is goal
   c) Various results based on how the approach

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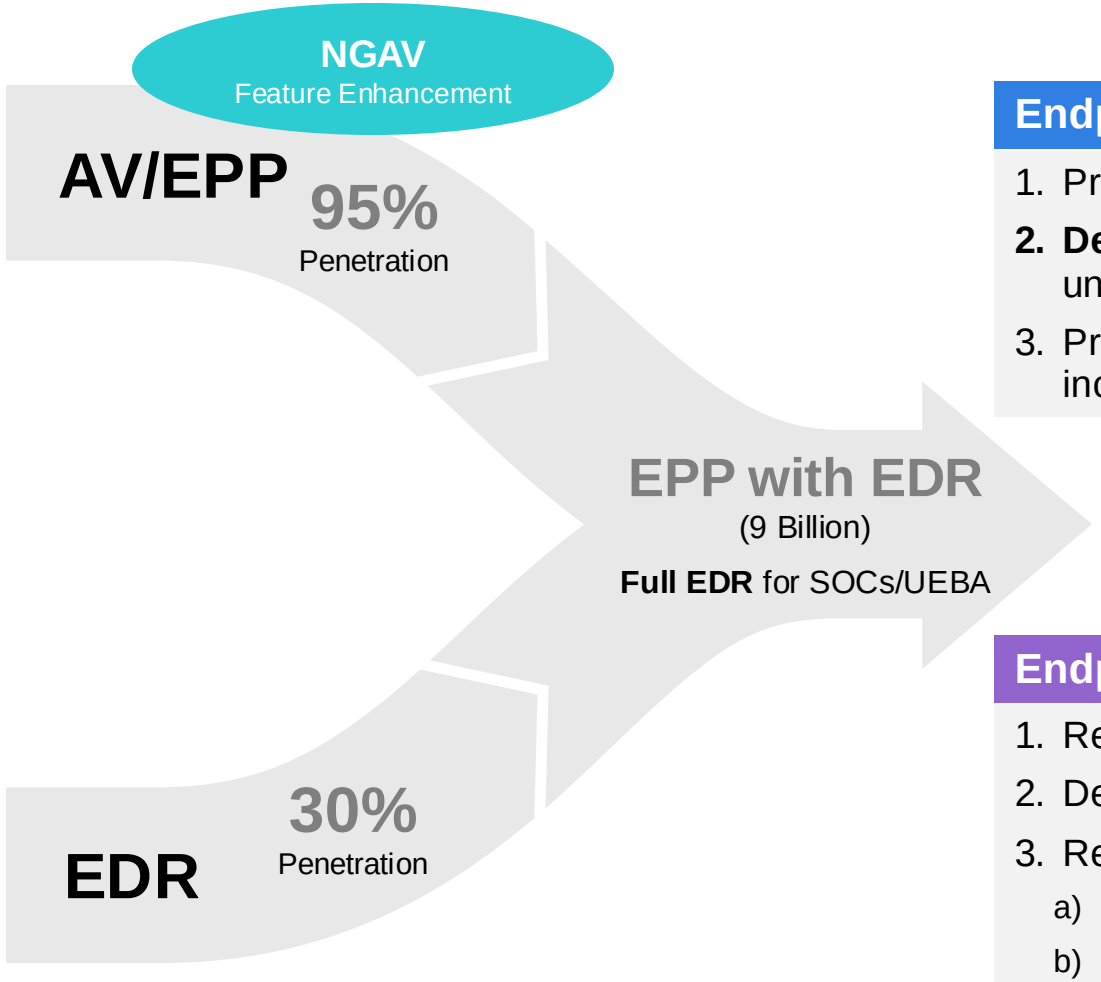
Evolution of Endpoint Security

Advanced Endpoint Protection Platform (EPP) with Automated Detection and Response Capabilities



Endpoint Protection Platform (EPP)

1. Prevent file-based malware
2. **Detect** and block malicious activities from *trusted* and untrusted applications
3. Provide basic capabilities to **respond** to security incidents and alerts

Endpoint Detection and Response (EDR)

1. Record and store endpoint behaviors and events
2. Detect security incidents
3. Response
   a) Containment and remediate
   b) Forensic—investigate security incidents, root cause analysis

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FortiEDR – Cloud Native EPP + EDR

Detect, Defuse, Respond and Remote Remediation

Pre-infection/ Pre-execution		Post-infection/Post-execution			
Discover	**Prevent**	**Detect**	**Defuse**	**Respond & Investigate**	**Remediate & Roll back**
Proactive risk mitigation	Pre-execution protection	File-less and advanced threats	Stop Breach and Ransomware	Full attack visibility	Automated Dis-infection
• Discover rogue devices & IoT • Vulnerabilities • Virtual patching	• Machine learning AV • FortiGuard Threat Intelligence • Kernel-level	• Behavioral based • Detect memory based attacks • Threat classification	• Block malicious actions • Prevent data loss • Zero Dwell time	• Playbook automation • Cross platform response • Forensic data • Behavioral-based threat hunting • Built-in MITRE tags	• Clean up / Roll back • Eliminate re-image/rebuild • Minimize business disruption

Automation | Cloud . Hybrid | OS coverage

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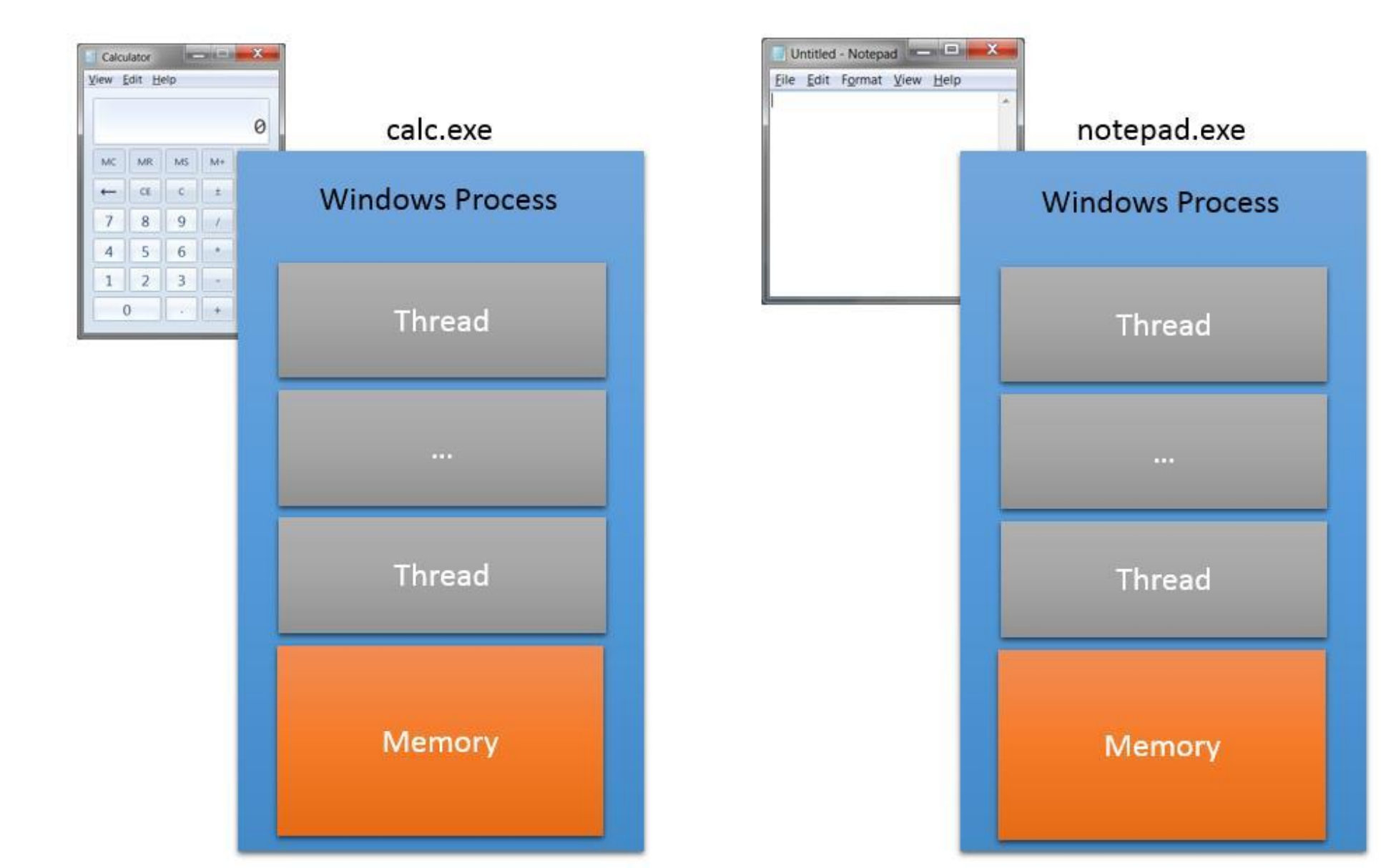

https://labs.nettitude.com/blog/dll-injection-part-one/

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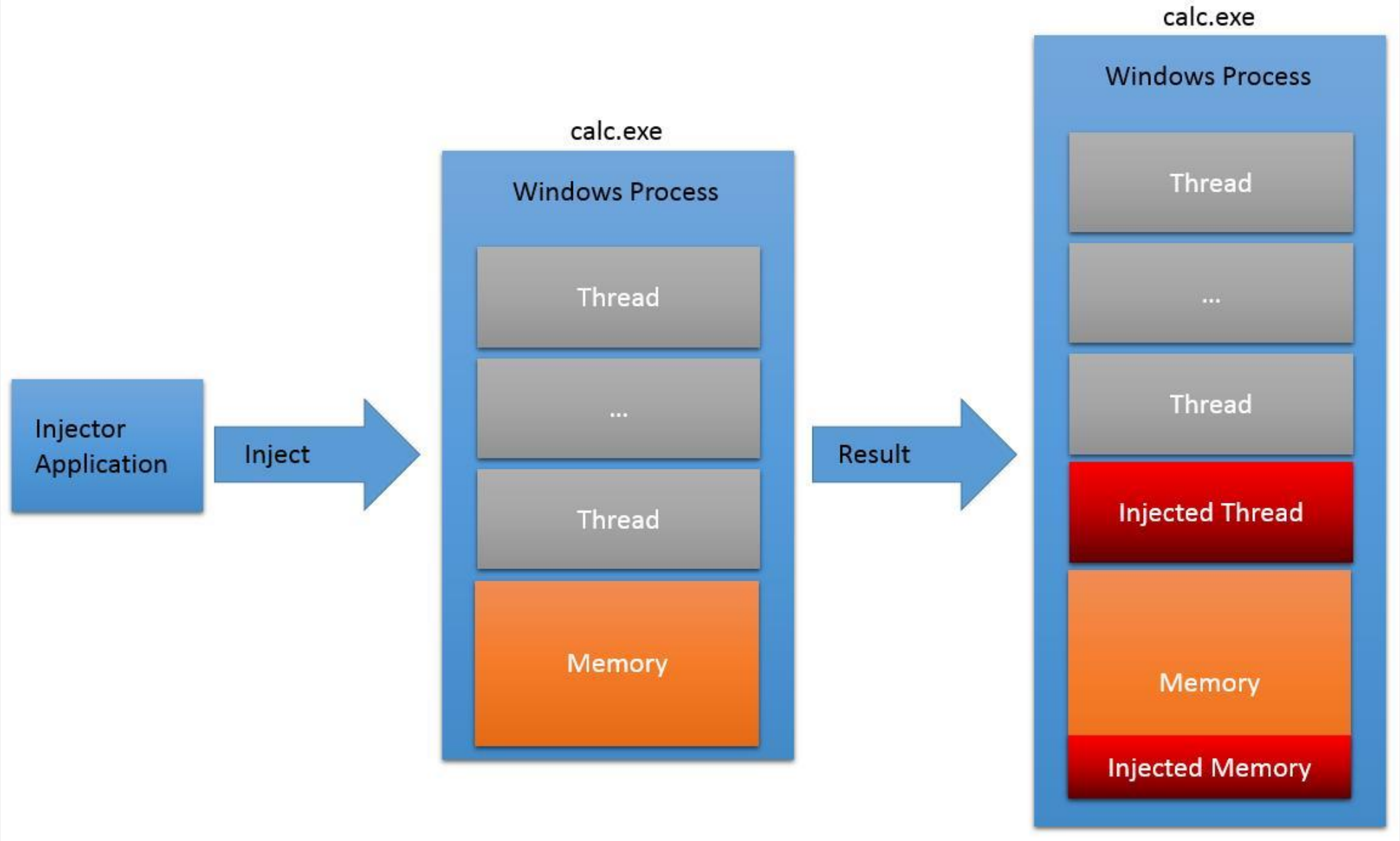

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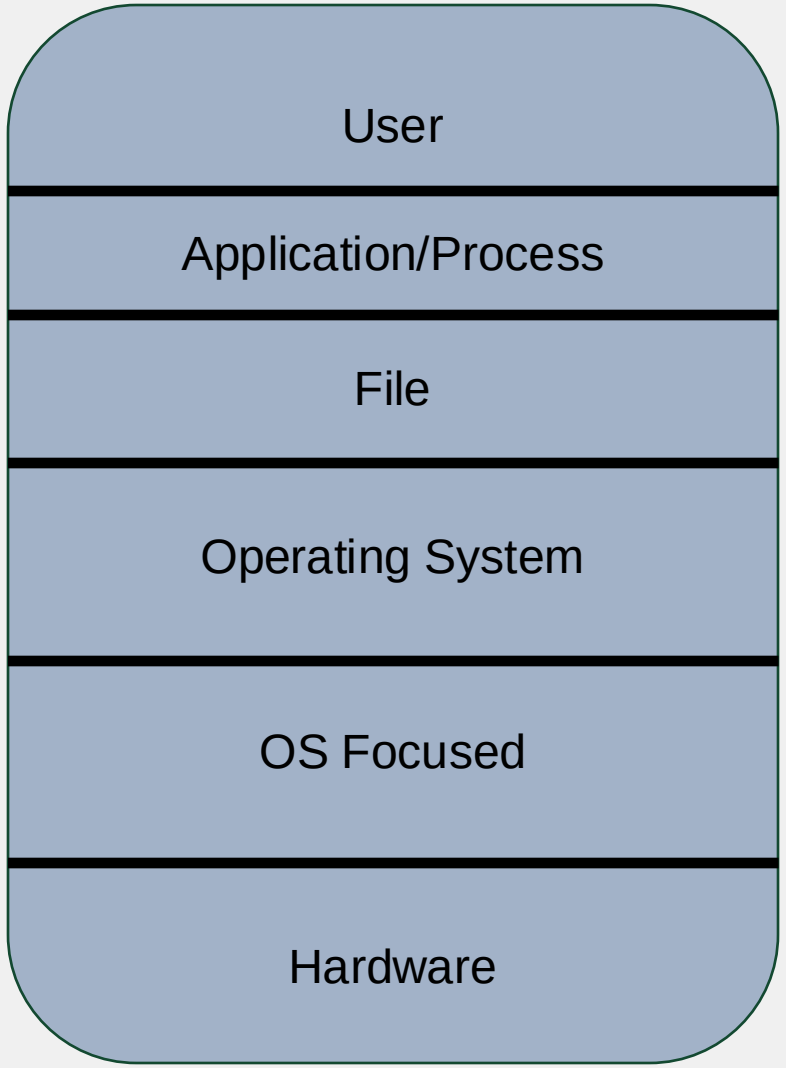
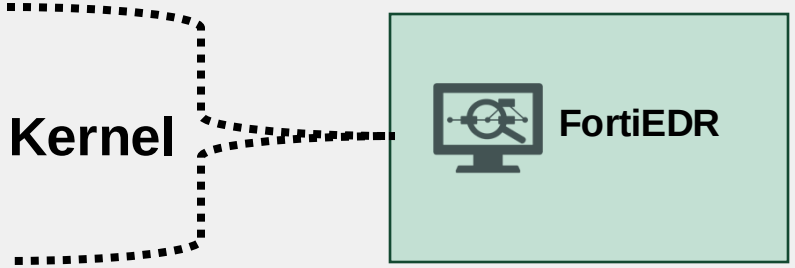
Kernel wins the day



Most other vendors…this eats up RAM.



OS Behavior Visibility
Low resource usage
Everything must go to the Kernel

*MacOS – Apple removed kernel extensions starting in BigSur, user space
*Linux – Most major distros are kernel, some are user space

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Automated Incident Response

AI Based Incident Response – *Why*?

Skills Gap

Huge shortage in cybersecurity personnel

- 2.72M cyber skills gap (ISC)
- At small and mid-size organizations IR is often an IT role

Response Time

Response time is critical

- Ransomware is the most obvious reason
- Machines are far better in sifting through large amounts of data

Complexity

Investigation and Response are far beyond “Quarantine File”

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File Rollback

FortiEDR Patented File Rollback

1) Most other vendors use Windows Volume Shadow Copies (VSS)

2) This creates bloat on the laptop and eats up HDD space

3) You’re using a Windows feature for security….seriously?!

4) Threat actors are actively going after VSS.

5) FortiEDR rollback is automatic, built-in.

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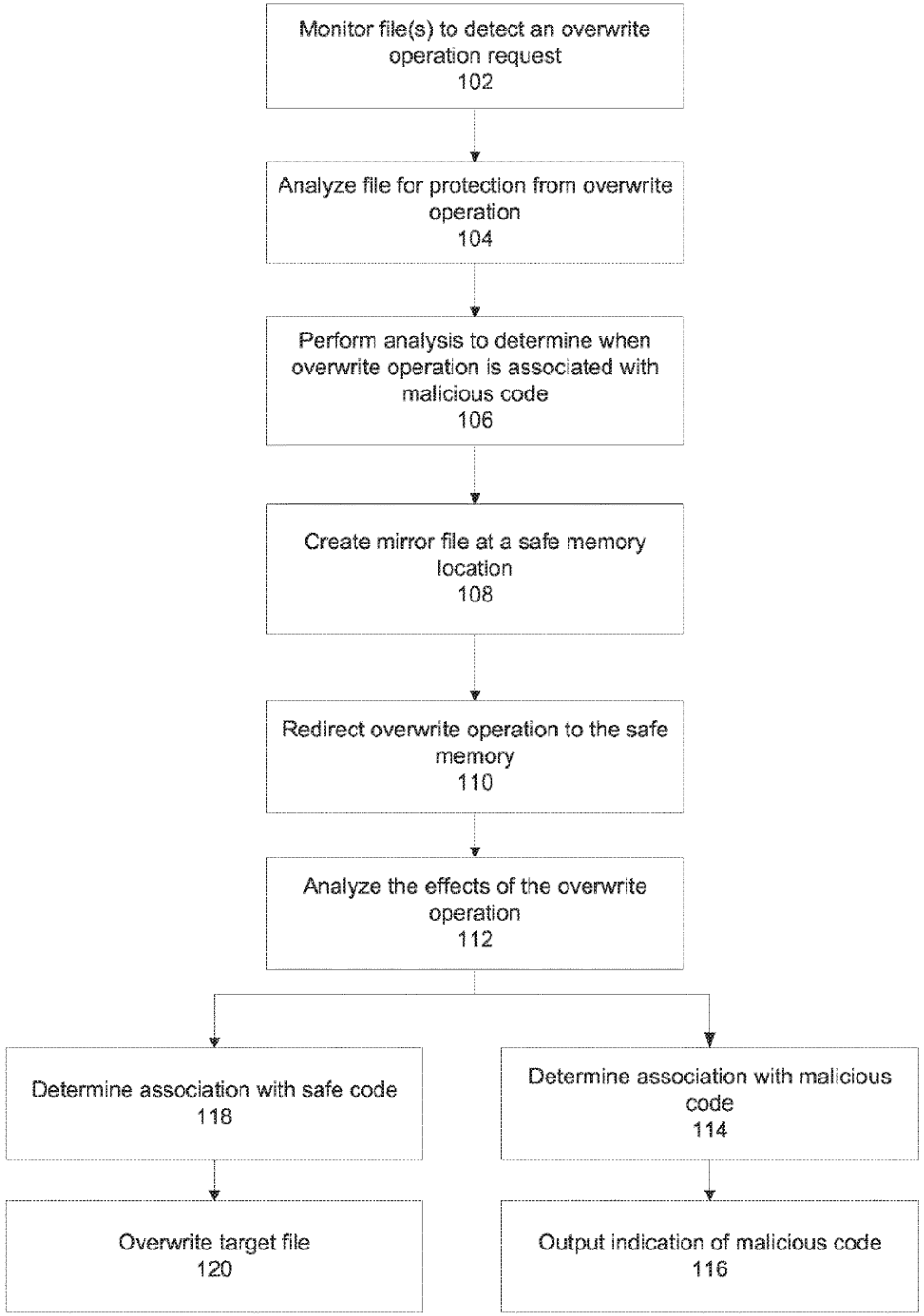
File Rollback Flow



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What is FortiEDR in summary

1) A lightweight agent, called a collector, running in kernel space Windows OS, some Linux.

2) Zero Trust Execution Model. Just because it ran clean yesterday doesn't meant we trust it today. Think of the SolarWinds hack.

3) Uses OS level rules to inspect all executions in the OS and automatic rollback if needed.

4) Has a Pre Execution (AKA) NextGen AV to stop common/known threats.

5) Uses patented technology called Post Execution Prevention to stop Zero Days.

6) Automated Incident Response to remove manual processes for SOC/Cyber Teams. Response happens in real-time.

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Anatomy of an Attack

Walk through of well-known attacks

Darkside Ransomware

- https://community.fortinet.com/t5/FortiEDR/Threat-Coverage-How-FortiEDR-detects-and-protects-against/ta-p/190322

Sunburst – Supply Chain Attack

- https://www.fortinet.com/blog/threat-research/what-we-have-learned-so-far-about-the-sunburst-solarwinds-hack

Threat Coverage – Post Exploit

- https://community.fortinet.com/t5/FortiEDR/Threat-Coverage-FortiEDR-mitigates-the-risk-of-post-exploitation/ta-p/210312

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LAB TIME!!

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